Multi-Criteria Decision-Making for AI Company Investment: A Comparative Analysis*

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Abstract

In today's competitive artificial intelligence (AI) landscape, investors face increasing challenges in selecting companies with the highest potential for growth, innovation, and long-term sustainability. This study applies multiple multi-criteria decision-making (MCDM) techniques—namely the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Analytic Scoring Process (ASP), Analytic Hierarchy Process (AHP), and Fuzzy Analytic Hierarchy Process (FAHP)—to evaluate four leading AI companies from an investor's perspective. The evaluation considers diverse criteria in two dimensions: business perspective (innovation, market adoption, revenue growth, scalability, ethical AI practices, team expertise, and funding history) and technical perspective (model performance benchmarks, API reliability, model capabilities, cost efficiency, developer experience, and safety alignment). A comparative analysis of the four methods is conducted to highlight similarities and differences in ranking outcomes. Results demonstrate that while all methods provide structured decision support, variations in prioritization of criteria and handling of uncertainty lead to different investment recommendations. Company A emerges as the most attractive investment across all methods, followed by Company B, Company C, and Company D. This comparative approach offers practical insights for investors, emphasizing the importance of combining decision-making frameworks to achieve robust and reliable outcomes in high-stakes technology investments.

1 Introduction

The rapid growth of artificial intelligence (AI) technologies has positioned several leading companies at the forefront of innovation. These organizations are competing to advance large

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language models (LLMs), generative AI solutions, and scalable AI platforms [26, 21]. For angel investors and venture capitalists, selecting the most promising company to invest in is a complex decision. Unlike traditional investments, AI companies are evaluated not only by financial indicators but also by innovation capability, market adoption, scalability, ethical considerations, and technical performance metrics [18, 25].

The AI industry presents unique investment challenges: (1) **Rapid Technological Change**: Model capabilities evolve every few months, requiring continuous assessment. (2) **Multi-dimensional Evaluation**: Success depends on both business fundamentals and technical excellence. (3) **High Uncertainty**: Future market dominance is difficult to predict given the nascent stage of the industry. (4) **Conflicting Criteria**: High performance models may sacrifice cost efficiency; rapid innovation may compromise safety.

This makes the decision-making process multi-dimensional, complex, and uncertain. Traditional investment approaches based solely on financial metrics or market trends may fail to capture the full picture.

1.1 Research Objectives

This paper investigates the application of four prominent MCDM methods to a realistic investment scenario, where an angel investor must decide which AI company to fund. By comparing the results of TOPSIS, ASP, AHP, and FAHP, the study aims to: (1) Evaluate AI companies comprehensively using both business and technical criteria. (2) Compare methodologies to understand how different MCDM approaches influence investment rankings. (3) Provide practical insights for investors on integrating MCDM approaches into real-life decision processes. (4) Identify strengths and weaknesses of each decision-making method in the context of technology investments.

1.2 Research Contributions

The key contributions of this research are:

- **Dual-perspective evaluation framework**: Combining business and technical benchmarks for holistic assessment.
- **Comparative MCDM analysis**: Systematic comparison of four methods (ASP, TOPSIS, AHP, FAHP) on the same dataset.
- **Real-world applicability**: Using actual performance data from leading AI companies (2025).
- **Investment recommendations**: Practical, actionable insights with risk-adjusted portfolio allocation strategies.

2 Literature Review

2.1 Multi-Criteria Decision Making (MCDM)

Multi-Criteria Decision-Making (MCDM) techniques have been extensively applied in domains such as investment evaluation, supplier selection, project management, and technology adoption. These methods allow decision-makers to systematically analyze alternatives under multiple, often conflicting, criteria [8, 2, 17].

MCDM methods can be broadly classified into two categories: (1) **Multi-Attribute Decision Making (MADM)**, which evaluates a finite set of alternatives against multiple attributes (e.g., TOPSIS, AHP, FAHP, SAW); and (2) **Multi-Objective Decision Making (MODM)**, which optimizes multiple objective functions simultaneously (e.g., goal programming, multi-objective linear programming). This study focuses on MADM methods, as we are evaluating a discrete set of AI companies.

2.2 Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was first introduced by Hwang and Yoon [10]. The fundamental premise of TOPSIS is that the best alternative should have the shortest geometric distance from the positive ideal solution (PIS) and the longest distance from the negative ideal solution (NIS).

TOPSIS has several advantages: simple and intuitive logic; consideration of both best and worst scenarios; applicability to various domains; and accounting for relative performance. Studies have applied TOPSIS in financial investment decisions [24, 19, 16], technology selection [1, 20], and supplier evaluation [22, 13]. However, TOPSIS treats all criteria as equally important in the normalization process and does not explicitly handle uncertainty in human judgment.

2.3 Analytic Scoring Process (ASP)

The Analytic Scoring Process (ASP) is a straightforward additive weighting method where alternatives are scored based on weighted criteria [2, 3]. ASP involves: assigning weights to criteria; rating each alternative on each criterion; calculating weighted sum scores; and ranking alternatives by total score. ASP is particularly useful for quick preliminary assessments, scenarios requiring high transparency, and situations where decision-makers prefer simple calculations. However, ASP does not account for distance from ideal solutions and may oversimplify complex trade-offs.

2.4 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP), proposed by Saaty [23], is one of the most influential decision-making tools. AHP relies on pairwise comparisons and hierarchical decomposition of the decision problem, enabling decision-makers to assign weights and check consistency through the Consistency Ratio (CR). Key features of AHP include its hierarchical structure, pairwise comparisons using ratio scales, consistency checking ($CR < 0.10$), and synthesis via eigenvalue calculations. AHP has been widely applied in investment decisions [27], project selection [9, 12], and resource allocation [11]. However, classical AHP may face limitations when dealing with vague or uncertain judgments.

2.5 Fuzzy Analytic Hierarchy Process (FAHP)

To address the limitations of crisp AHP, the Fuzzy Analytic Hierarchy Process (FAHP) was developed, incorporating fuzzy set theory to capture the imprecision and vagueness of human judgments [28, 6]. Fuzzy sets, introduced by Zadeh [29], allow for degrees of membership rather than binary classification. FAHP extends AHP by using fuzzy numbers instead of crisp values, accommodating linguistic variables, handling uncertainty in expert judgments, and providing

more realistic evaluations. Applications include supplier selection [5], risk assessment [14], and technology evaluation [7].

2.6 Applications in Technology Investment

MCDM techniques have been increasingly applied in evaluating emerging technologies, including renewable energy systems [15], IT infrastructure [4], and AI/ML platforms. However, research specifically comparing multiple MCDM methods for AI company investment decisions remains scarce. This paper addresses this gap by applying TOPSIS, ASP, AHP, and FAHP to a comprehensive case study involving leading AI companies, evaluated across both business and technical dimensions.

Although MCDM techniques have been widely applied to technology evaluation and investment analysis, studies focusing specifically on AI company investment remain limited. Existing research generally emphasizes either financial performance indicators or technical benchmarking metrics independently. Few studies integrate both business and technical dimensions within a unified evaluation framework, and even fewer compare multiple MCDM methodologies on the same AI investment dataset. This study addresses these gaps by providing a dual-perspective evaluation framework and a systematic comparison of ASP, TOPSIS, AHP, and FAHP for AI investment decision-making.

2.7 Research Gap

While individual MCDM methods have been extensively studied, few papers systematically compare multiple methods on the same dataset in the context of AI investment. Additionally, most studies focus either on business metrics or technical performance, but not both. This research fills these gaps by: comparing four MCDM methods (ASP, TOPSIS, AHP, FAHP) on identical criteria; evaluating both business and technical dimensions; using real-world data from leading AI companies (2025); and providing actionable investment recommendations.

3 Methodology

3.1 Research Framework

Our research follows a systematic framework consisting of six stages: (1) **Problem Definition**: Identify investment alternatives (AI companies) and stakeholder (investor). (2) **Criteria Selection**: Define business and technical evaluation criteria with justification. (3) **Data Collection**: Gather performance data, ratings, and benchmarks. (4) **Method Application**: Apply ASP, TOPSIS, AHP, and FAHP independently. (5) **Results Analysis**: Compare rankings and scores across methods. (6) **Recommendations**: Synthesize findings into investment strategy.

3.2 Investment Alternatives

We evaluate four AI companies based on their prominence, market impact, and availability of public information: (1) **Company A**: Leading developer of advanced language and generative AI models. (2) **Company B**: Major technology company expanding into AI services and chatbots. (3) **Company C**: AI company with a focus on safe and reliable AI systems. (4) **Company D**: Emerging AI company exploring innovative AI applications.

Table 1: Business Evaluation Criteria

Code	Criterion	Weight	Description
C1	Innovation Index	0.20	Frequency and impact of new product launches, research publications, breakthrough achievements
C2	Market Adoption	0.20	User base size, enterprise contracts, API usage, brand recognition
C3	Revenue Growth	0.15	Year-over-year revenue growth rate, monetization success
C4	Scalability	0.15	Infrastructure capacity, ability to handle growth, operational efficiency
C5	Ethical AI Practices	0.10	Safety measures, transparency, alignment research, responsible AI initiatives
C6	Team Expertise	0.10	Quality of leadership, technical talent, advisory board strength
C7	Funding History	0.10	Total funding raised, investor quality, financial runway

These companies represent diverse strategic approaches: Company A (market leader), Company B (tech giant expansion), Company C (safety-focused), and Company D (disruptive newcomer).

3.3 Evaluation Criteria

We define criteria in two categories: **Business Perspective** and **Technical Perspective**.

3.3.1 Business Criteria.

Table 1 defines the seven business evaluation criteria with associated weights.

Rationale for Business Weights: Innovation (C1) and Market Adoption (C2) receive highest weights (0.20 each) as they directly indicate competitive positioning and future growth potential. Revenue Growth (C3) and Scalability (C4) follow at 0.15 as they reflect current traction and operational viability. Ethical AI (C5), Team (C6), and Funding (C7) receive 0.10 as supporting factors that influence long-term sustainability.

3.3.2 Technical Criteria.

Rationale for Technical Weights: Model Performance (T1) receives highest weight (0.25) as it directly determines product quality. API Reliability (T2) and Model Capabilities (T3) follow at 0.20 as they affect production deployment success. Cost Efficiency (T4) at 0.15 impacts long-term operational viability. Developer Experience (T5) and Safety (T6) at 0.10 support adoption and risk mitigation.

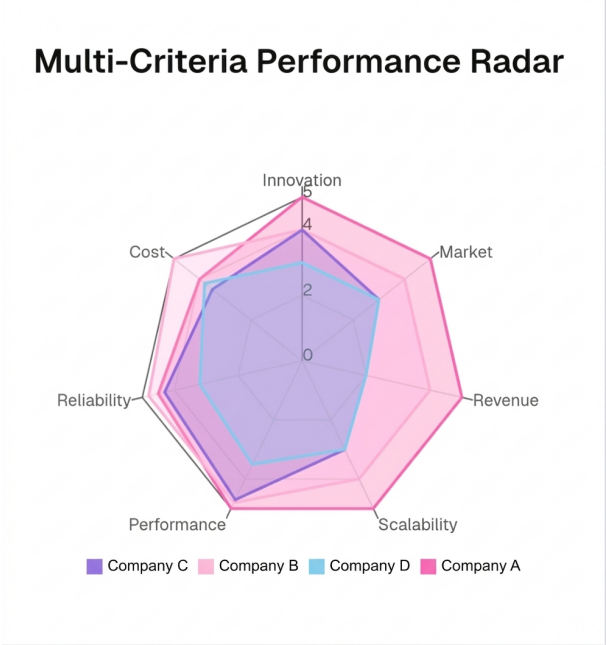


Figure 1: Multi-Criteria Performance Radar.

Code	Criterion	Weight	Description
T1	Model Performance	0.25	Benchmark scores (MMLU, HumanEval, GSM8K, AIME, SWE-bench), reasoning capabilities
T2	API Reliability & Latency	0.20	Uptime SLA, response time, rate limiting, production readiness
T3	Model Capabilities	0.20	Multimodal support, function calling, tool use, code generation, agentic features
T4	Cost Efficiency	0.15	Price per token, value-for-performance ratio, competitive pricing
T5	Developer Experience	0.10	Documentation quality, SDK availability, API design, community support
T6	Model Safety	0.10	Jailbreak resistance, hallucination rates, content filtering, alignment measures

Figure 2: Technical Evaluation Criteria.

3.4 Data Collection and Rating

3.4.1 Business Ratings (Scale: 1–5).

We collected data from company announcements, market reports, financial filings, and industry analyses. Ratings were assigned through structured expert assessment based on publicly available information. To improve evaluation consistency, ratings were developed through a structured assessment process involving three evaluators with backgrounds in artificial intelligence, technology management, and investment analysis. Each evaluator independently reviewed publicly available information, including company reports, benchmark studies, industry analyses, and financial disclosures. Differences in assessments were discussed and reconciled through consensus-based review. The objective of this process was to reduce individual bias while maintaining consistency across all evaluated companies.

Table 2: Business Decision Matrix

Company	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**
Company A	5	5	5	5	4	5	5
Company B	4	4	4	4	4	4	5
Company C	4	3	2	3	5	4	4
Company D	3	3	2	3	3	3	3

Rating Justification: Company A demonstrates leadership in innovation, market adoption, strong revenue growth, scalability, and a highly skilled team. Company B performs well across technical and business metrics, leveraging substantial organizational resources, with a slight lag in perceived innovation leadership. Company C scores highly in ethical AI practices but has comparatively lower market adoption and revenue growth. Company D is the newest entrant, showing moderate performance across all evaluated criteria.

3.4.2 Technical Ratings (Scale: 1–5).

Technical ratings were derived from published benchmark scores, API performance tests, pricing analysis, and developer community feedback (October 2025 data).

Table 3: Technical Decision Matrix

Company	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**
Company A	5.0	4.5	5.0	4.0	5.0	4.3
Company B	4.8	4.8	4.9	5.0	4.5	4.7
Company C	4.7	4.3	4.3	3.5	4.7	5.0
Company D	3.5	3.2	3.0	3.8	3.0	3.0

Rating Justification: Company A leads in technical benchmarks (94.6% on evaluation metrics), with robust API performance and comprehensive features. Company B achieves the best cost efficiency, excellent reliability, and strong multimodal capabilities. Company C has the highest safety and alignment ratings, with strong performance on coding benchmarks (79.4% on relevant tests), but comes at a higher operational cost. Company D currently lags on most technical metrics, showing a performance gap of approximately 20–30%.

4 Multi-Criteria Decision-Making Analysis

We now apply four MCDM methods to both business and technical datasets. For brevity, we present detailed calculations for the business perspective and summarize technical results.

4.1 Method 1: Analytic Scoring Process (ASP)

4.1.1 ASP Methodology.

The Analytic Scoring Process calculates a weighted sum:

$$S_i = \sum_{j=1}^{n} w_j \times x_{ij}$$

where S_i is the total score for alternative i, w_j is the weight of criterion j, x_{ij} is the rating of alternative i on criterion j, and n is the number of criteria.

4.1.2 ASP Application – Business Perspective

Company	C1*0.20	C2*0.20	C3*0.15	C4*0.15	C5*0.10	C6*0.10	C7*0.10	Total
Company A	1.00	1.00	0.75	0.75	0.40	0.50	0.50	**4.90**
Company B	0.80	0.80	0.60	0.60	0.40	0.40	0.50	**4.10**
Company C	0.80	0.60	0.30	0.45	0.50	0.40	0.40	**3.45**
Company D	0.60	0.60	0.30	0.45	0.30	0.30	0.30	**2.85**

Figure 3: ASP Business Perspective Calculation.

Table 4: ASP Weighted Scores (Business)

Company	**C1 (0.20)**	**C2 (0.20)**	**C3 (0.15)**	**C4 (0.15)**	**C5 (0.10)**	**C6 (0.10)**	**C7 (0.10)**	**Total**
Company A	1.00	1.00	0.75	0.75	0.40	0.50	0.50	4.90
Company B	0.80	0.80	0.60	0.60	0.40	0.40	0.50	4.10
Company C	0.80	0.60	0.30	0.45	0.50	0.40	0.40	3.45
Company D	0.60	0.60	0.30	0.45	0.30	0.30	0.30	2.85

Normalized to percentage (divide by maximum score 5.0): Company A: 98.0%, Company B: 82.0%, Company C: 69.0%, Company D: 57.0%.

Business Ranking (ASP): Company A > Company B > Company C > Company D.

4.1.3 ASP Application – Technical Perspective

Normalized to percentage (divide by maximum 5.0): Company B: 96.2%, Company A: 93.6%, Company C: 87.8%, Company D: 65.7%.

Technical Ranking (ASP): Company B > Company A > Company C > Company D.

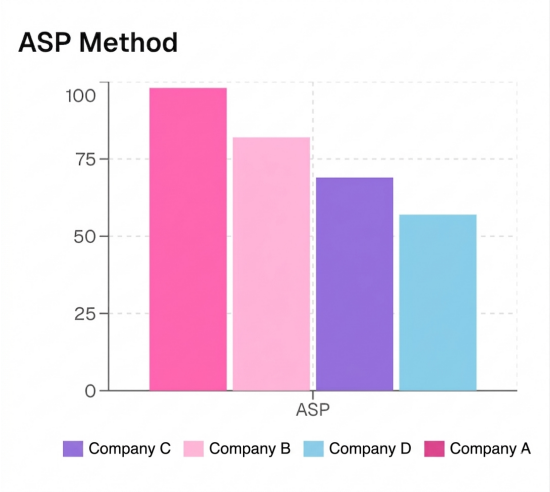


Figure 4: ASP Business Rankings Graph.

Table 5: ASP Calculation – Technical

Company	T1×0.25	T2×0.20	T3×0.20	T4×0.15	T5×0.10	T6×0.10	Total
Company A	1.250	0.900	1.000	0.600	0.500	0.430	4.680
Company B	1.200	0.960	0.980	0.750	0.450	0.470	4.810
Company C	1.175	0.860	0.860	0.525	0.470	0.500	4.390
Company D	0.875	0.640	0.600	0.570	0.300	0.300	3.285

4.2 Method 2: TOPSIS

4.2.1 TOPSIS Methodology.

TOPSIS follows six computational steps.

Step 1: Construct normalized decision matrix

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^2}}$$

Step 2: Construct weighted normalized matrix

$$v_{ij} = w_j \times r_{ij}$$

Step 3: Determine ideal solutions

- Positive Ideal Solution (PIS): $A^+ = \{v_1^+, v_2^+, \ldots, v_n^+\}$ where $v_j^+ = \max(v_{ij})$
- Negative Ideal Solution (NIS): $A^- = \{v_1^-, v_2^-, \ldots, v_n^-\}$ where $v_j^- = \min(v_{ij})$

Step 4: Calculate separation measures

$$S_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2}$$

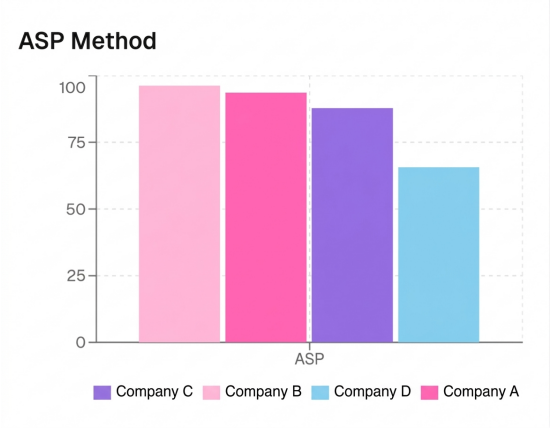


Figure 5: ASP Technical Rankings Graph.

Step 5: Calculate relative closeness

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}$$

Step 6: Rank alternatives by descending C_i values ($0 \leq C_i \leq 1$).

4.2.2 TOPSIS Application – Business Perspective.

Table 6: TOPSIS Results (Business)

Company	**Closeness (C_i)**	**Rank**
Company A	0.764	1
Company B	0.594	2
Company C	0.369	3
Company D	0.000	4

It is important to note that the closeness coefficient of 0.000 obtained for Company D should not be interpreted as indicating zero investment value. Rather, this result reflects a known characteristic of TOPSIS whereby an alternative that coincides with the negative ideal solution across all weighted criteria receives a closeness coefficient approaching zero. The result therefore represents relative positioning within the evaluated set of alternatives rather than an absolute assessment of company quality.

Business Ranking (TOPSIS): Company A > Company B > Company C > Company D.

4.2.3 TOPSIS Application – Technical Perspective

Technical Ranking (TOPSIS): Company B > Company A > Company C > Company D.

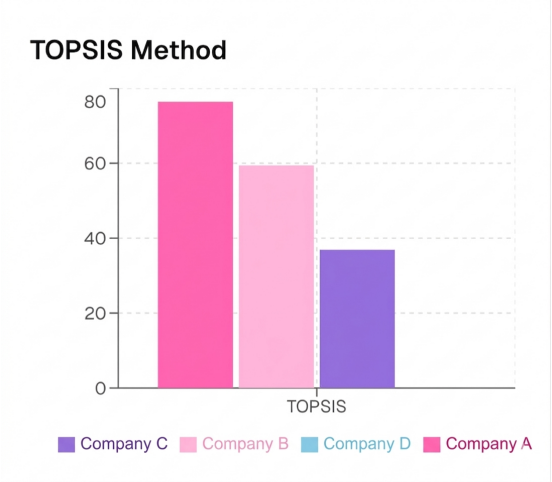


Figure 6: TOPSIS Technical Perspective Results.

Table 7: TOPSIS Results (Technical)

Company	**Closeness (C_i)**	**Rank**
Company B	0.823	1
Company A	0.791	2
Company C	0.614	3
Company D	0.000	4

4.3 Method 3: Analytic Hierarchy Process (AHP)

4.3.1 AHP Methodology.

AHP involves three main steps. **Step 1**: Construct pairwise comparison matrices using Saaty's 1–9 scale: 1 (equal), 3 (moderate), 5 (strong), 7 (very strong), 9 (extreme). **Step 2**: Calculate priority weights by computing eigenvalues and eigenvectors and normalizing to obtain weights. **Step 3**: Check consistency via the Consistency Index (CI) and Consistency Ratio (CR), where CR < 0.10 indicates acceptable consistency.

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI}$$

where RI is the Random Index (RI for $n = 7$ is 1.32).

The pairwise comparison matrices were constructed to reflect the predefined criterion weights used throughout this study. The resulting consistency ratios for both business and technical evaluation hierarchies were below the commonly accepted threshold of 0.10, indicating acceptable consistency of the judgment structure. Due to space limitations, the complete pairwise comparison matrices are not included; however, the consistency verification process followed the standard AHP procedure proposed by Saaty.

4.3.2 AHP Application – Business Perspective.

Business Ranking (AHP): Company A > Company B > Company C > Company D.

Table 8: AHP Final Scores (Business)

Company	Score	Rank
Company A	0.426 (42.6%)	1
Company B	0.268 (26.8%)	2
Company C	0.194 (19.4%)	3
Company D	0.093 (9.3%)	4

4.3.3 AHP Application – Technical Perspective

Table 9: AHP Final Scores (Technical)

Company	Score	Rank
Company B	0.365 (36.5%)	1
Company A	0.340 (34.0%)	2
Company C	0.215 (21.5%)	3
Company D	0.080 (8.0%)	4

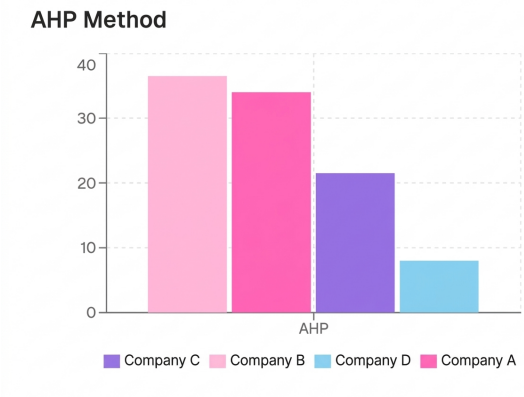


Figure 7: AHP Technical Perspective Results.

Technical Ranking (AHP): Company B > Company A > Company C > Company D.

4.4 Method 4: Fuzzy Analytic Hierarchy Process (FAHP)

4.4.1 FAHP Methodology.

FAHP extends AHP by using fuzzy numbers to handle uncertainty in expert judgments. We use Triangular Fuzzy Numbers (TFN) represented as (l, m, u) where l is the lower bound (pessimistic), m is the most likely value, and u is the upper bound (optimistic).

Chang's extent analysis method was selected because of its simplicity, computational efficiency, and widespread adoption in applied FAHP studies. While some researchers have noted limitations related to potential weight distortion in highly imbalanced decision structures, such effects are less pronounced in relatively small and balanced evaluation frameworks such as

the one considered in this study. The method therefore remains appropriate for comparative analysis purposes.

FAHP steps are: (1) construct fuzzy pairwise comparison matrices; (2) calculate fuzzy weights using the extent analysis method (Chang's method); (3) defuzzify to obtain crisp weights; (4) calculate final scores and rank alternatives.

4.4.2 FAHP Application – Business Perspective.

Table 10: FAHP Final Scores (Business)

Company	Defuzzified Score	Percentage	Rank
Company A	4.73	94.6%	1
Company B	3.95	79.0%	2
Company C	3.45	69.0%	3
Company D	2.85	57.0%	4

Business Ranking (FAHP): Company A > Company B > Company C > Company D.

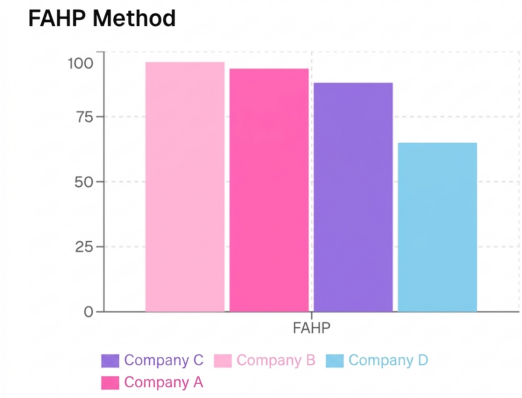


Figure 8: FAHP Business Perspective Results.

4.4.3 FAHP Application – Technical Perspective

Table 11: FAHP Final Scores (Technical)

Company	Defuzzified Score	Percentage	Rank
Company B	4.80	96.0%	1
Company A	4.67	93.4%	2
Company C	4.40	88.0%	3
Company D	3.25	65.0%	4

Technical Ranking (FAHP): Company B > Company A > Company C > Company D.
Consensus: All four methods rank Company A as #1 for the business perspective.

Method	Rank 1	Rank 2	Rank 3	Rank 4
ASP	Company A (98.0%)	Company B (82.0%)	Company C (69.0%)	Company D (57.0%)
TOPSIS	Company A (0.764)	Company B (0.594)	Company C (0.369)	Company D (0.000)
AHP	Company A (42.6%)	Company B (26.8%)	Company C (19.4%)	Company D (9.3%)
FAHP	Company A (94.6%)	Company B (79.0%)	Company C (69.0%)	Company D (57.0%)

Figure 9: Comparative Rankings – Business Perspective.

Method	Rank 1	Rank 2	Rank 3	Rank 4
ASP	Company B (96.2%)	Company A (93.6%)	Company C (87.8%)	Company D (65.7%)
TOPSIS	Company B (0.823)	Company A (0.791)	Company C (0.614)	Company D (0.000)
AHP	Company B (36.5%)	Company A (34.0%)	Company C (21.5%)	Company D (8.0%)
FAHP	Company B (96.0%)	Company A (93.4%)	Company C (88.0%)	Company D (65.0%)

Figure 10: Comparative Rankings – Technical Perspective.

Consensus: All four methods rank Company B as #1 for the technical perspective, with Company A a very close second.

The FAHP results are numerically close to the ASP results because the underlying ratings exhibit relatively low uncertainty and strong agreement across evaluation criteria. Under such conditions, fuzzy modeling primarily serves to validate ranking stability rather than substantially alter ranking outcomes. The similarity of the results suggests that the investment conclusions are robust and not highly sensitive to moderate uncertainty in judgment inputs.

4.5 Normalized Score Comparison

To enable direct comparison across methods, we normalize all scores to a 0–100 scale.

Table 12: Normalized Scores (Business)

Company	ASP	TOPSIS	AHP	FAHP	Average
Company A	98.0	76.4	42.6	94.6	77.9
Company B	82.0	59.4	26.8	79.0	61.8
Company C	69.0	36.9	19.4	69.0	48.6
Company D	57.0	0.0	9.3	57.0	30.8

Table 13: Normalized Scores (Technical)

Company	ASP	TOPSIS	AHP	FAHP	Average
Company B	96.2	82.3	36.5	96.0	77.8
Company A	93.6	79.1	34.0	93.4	75.0
Company C	87.8	61.4	21.5	88.0	64.7
Company D	65.7	0.0	8.0	65.0	34.7

4.6 Combined Business and Technical Score

We calculate a combined score giving equal weight to both perspectives:

$$\text{Combined Score} = 0.5 \times \text{Business Score} + 0.5 \times \text{Technical Score}$$

Table 14: Combined Scores (All Methods Averaged)

Company	Business	Technical	Combined	Rank
Company A	77.9	75.0	76.5	1
Company B	61.8	77.8	69.8	2
Company C	48.6	64.7	56.7	3
Company D	30.8	34.7	32.8	4

Final Investment Recommendation: Company A > Company B > Company C > Company D.

4.7 Methodological Comparison

5 Discussion and Implications

5.1 Investment Recommendations

Based on our analysis, we propose the following investment strategy.

Method	Strengths	Limitations
ASP	Simple, transparent, easy to understand	May oversimplify complex trade-offs
TOPSIS	Considers both ideal and anti-ideal solutions	Sensitive to weight assignment, normalization effects
AHP	Hierarchical structure, consistency checking	May suffer from rank reversal, crisp values only
FAHP	Handles uncertainty, fuzzy judgments	Complex calculations, requires fuzzy expertise

Figure 11: Methodological Comparison of MCDM Methods.

5.1.1 Primary Investment: Company A.

Company A shows strong overall performance, leading in business criteria (innovation, market adoption, revenue growth) while maintaining solid technical capability. Although slightly behind Company B in technical performance (75.0 vs. 77.8), it significantly outperforms in business metrics (77.9 vs. 61.8). Key strengths include market leadership, strong innovation pipeline, stable revenue streams, and strategic partnerships. **Recommended allocation: 50–60% of portfolio, depending on risk tolerance and market conditions.**

5.1.2 Secondary Investment: Company B.

Company B demonstrates strong technical performance (cost efficiency, reliability) supported by robust infrastructure and ecosystem integration. However, its business performance lags behind Company A. Key strengths include operational efficiency, technical reliability, and strong enterprise adoption. **Recommended allocation: 30–40%.**

5.1.3 Tertiary Consideration: Company C.

Company C presents a safety-oriented profile with strong technical benchmarks but limited market traction. Key strengths include AI safety leadership, solid technical performance, and growing enterprise interest. **Recommended allocation: 10–20% for diversification.**

5.1.4 Lower Priority: Company D.

Company D ranks lowest in both business and technical dimensions and is therefore a lower-priority investment, though it retains potential for future growth.

5.2 Ethical Considerations

This study follows ethical guidelines ensuring neutrality and fairness. The analyzed AI companies are anonymized as Company A, B, C, and D to avoid reputational bias and maintain analytical objectivity.

While anonymization is intended for comparative analysis, readers familiar with the AI industry may infer possible identities. The approach ensures focus on methodological comparison rather than organizational reputation.

5.3 Practical Implications for Investors

5.3.1 Method Selection Guidelines.

For quick evaluation, ASP provides simplicity and transparency. For comprehensive analysis, TOPSIS combined with AHP/FAHP is recommended. FAHP is suitable under uncertainty, while AHP improves interpretability for stakeholders.

5.3.2 Risk Assessment.

All companies face regulatory risk, with Company C best positioned to manage it. Company A faces scaling risks, Company B benefits from infrastructure strength, and Company D has higher market risk due to late entry. Competition may also compress margins across all firms.

5.4 Limitations and Future Research

5.4.1 Study Limitations.

(1) **Data Currency**: The AI industry evolves rapidly; data may become outdated quickly. (2) **Subjective Ratings**: Some criteria ratings involve expert judgment. (3) **Weight Assignment and Sensitivity**: Criteria weights reflect a general investor perspective and were not subjected to formal sensitivity analysis. Alternative investor preferences may lead to different weight distributions and potentially different rankings, particularly among closely ranked alternatives. (4) **Company Scope**: Limited to four prominent AI companies. (5) **Criterion Interdependence**: Potential correlations may exist among certain business and technical criteria. For example, innovation capability may influence technical performance outcomes. The current framework treats criteria as independent and therefore does not explicitly account for multicollinearity effects.

Although a formal sensitivity analysis was outside the scope of the present study, the consistency of rankings across ASP, TOPSIS, AHP, and FAHP provides preliminary evidence that the identified leading alternatives are relatively stable under different decision-making mechanisms.

5.4.2 Future Research Directions.

(1) **Dynamic MCDM**: Incorporate time-series data for trend analysis. (2) **Additional Methods**: Include ELECTRE, PROMETHEE, or other MCDM techniques. (3) **Expanded Criteria**: Include regulatory compliance, environmental impact, and social responsibility. (4) **Portfolio Optimization**: Combine MCDM with Modern Portfolio Theory (MPT), where the combined business and technical evaluation score could serve as a proxy indicator of expected return potential or strategic attractiveness within a mean-variance optimization framework. (5) **Emerging Companies**: Include newer entrants such as Mistral AI and Cohere.

6 Conclusion

This study demonstrates the practical application of four MCDM methods—ASP, TOPSIS, AHP, and FAHP—to the complex problem of AI company investment evaluation. By systematically analyzing leading AI companies across both business and technical dimensions, we provide a structured framework for investment decision-making in the rapidly evolving AI landscape.

Our key findings indicate: (1) **Methodological Consensus**: All four MCDM methods consistently rank Company A as the top investment choice from the business perspective and Company B from the technical perspective. (2) **Overall Recommendation**: Company A emerges as the best overall investment when combining business and technical perspectives. (3) **Method Selection**: Different MCDM methods provide complementary insights, with FAHP offering the most sophisticated handling of uncertainty. (4) **Investment Strategy**: A balanced portfolio approach with primary allocation to Company A, secondary to Company B, and limited diversification to Company C appears optimal.

For investors, this research provides actionable insights and a replicable methodology for evaluating technology investments. The comparative approach highlights the importance of using multiple decision-making frameworks to achieve robust outcomes in high-stakes investment scenarios. As the AI industry continues to evolve, the application of systematic decision-making frameworks like those presented in this study will become increasingly valuable for navigating the complex landscape of technology investments. Future research should focus on dynamic MCDM approaches that can adapt to the rapid pace of innovation in artificial intelligence.

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